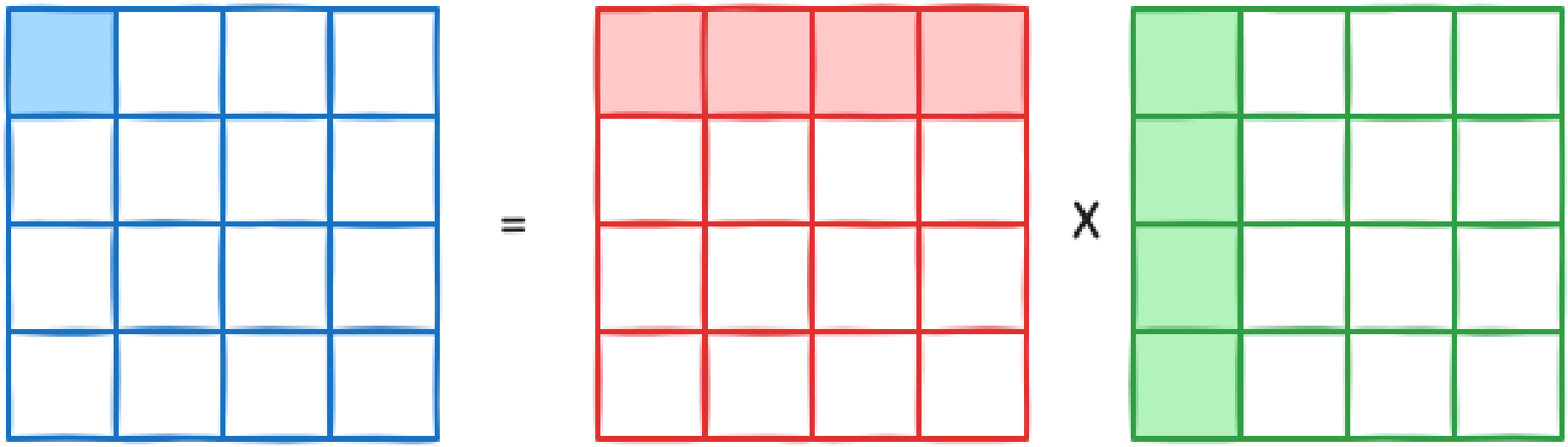
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1 Why GPUs? What's wrong with CPUs?

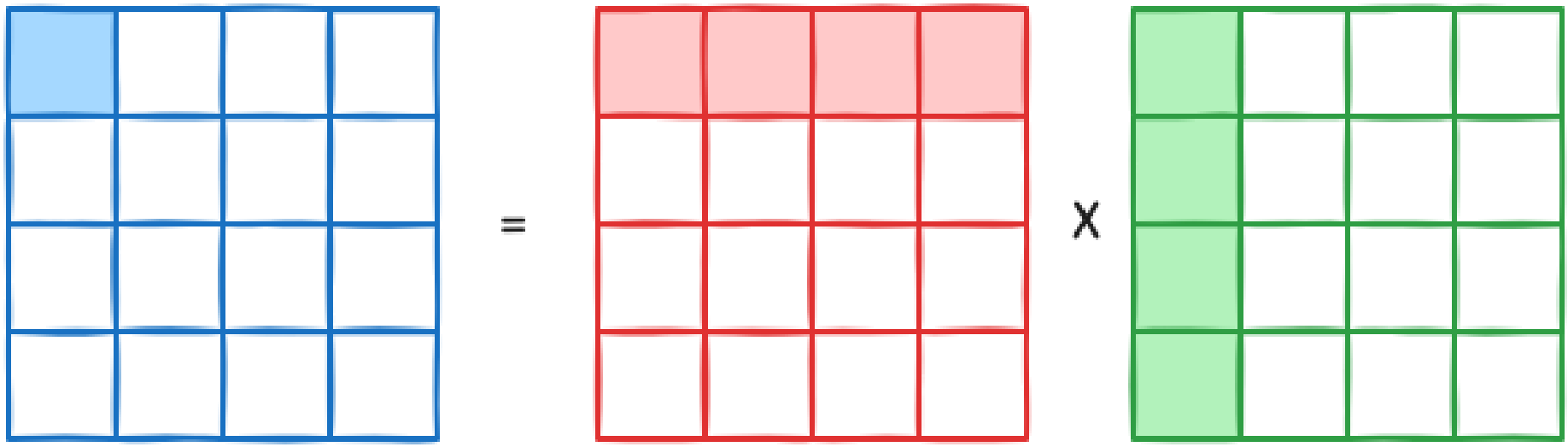
Computing a Matrix Multiplication

Let A and B be two 4×4 matrices. The matrix multiple $C = AB$ results in another 4×4 matrix.



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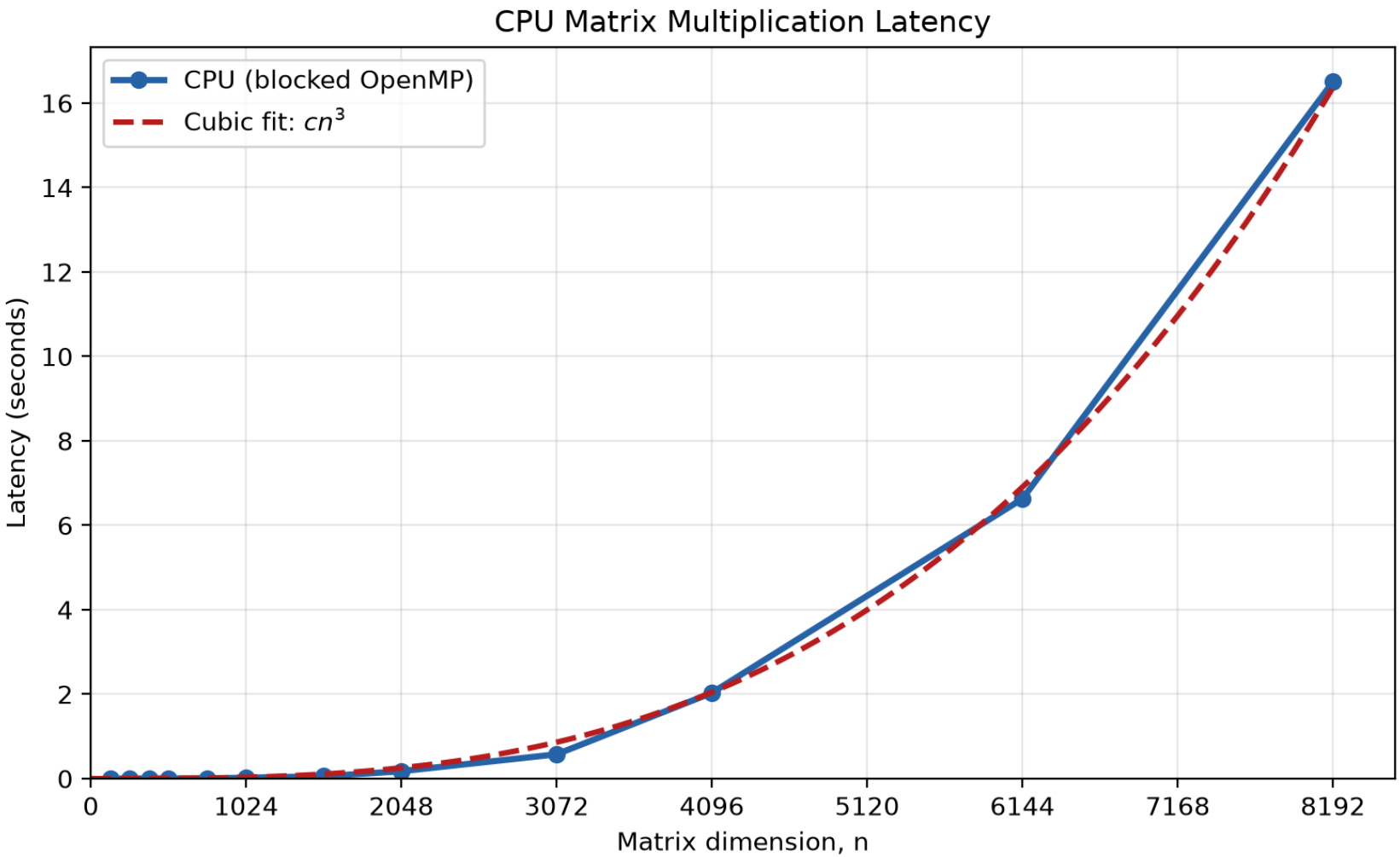
Poor scaling

The operations are very simple, but the number of operations grow on the order of $\mathcal{O}(n^3)$, where n is the dimensions of the matrices.

Runtime of a matrix multiplication on a CPU

```
for (std::size_t y = 0; y < n; ++y) {
  for (std::size_t x = 0; x < n; ++x) {
    double sum = 0.0;
    for (std::size_t k = 0; k < n; ++k) {
      sum += a[y*n+k] * b[k*n+x];
    }
    c[y*n+x] = sum;
  }
}
```

Runtime of a matrix multiplication on a CPU



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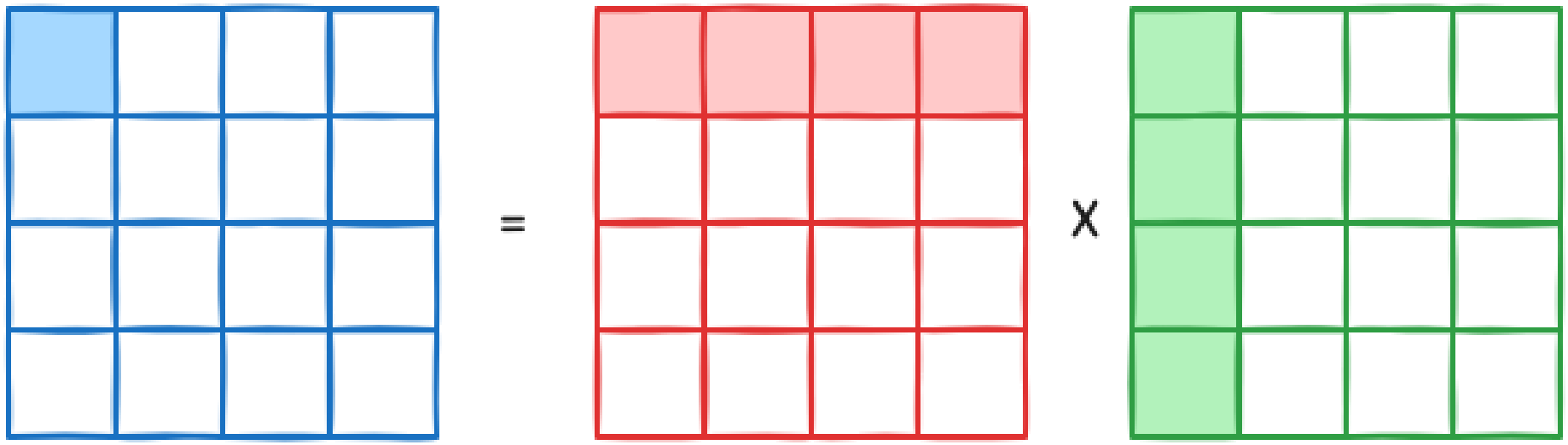
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The key tradeoff

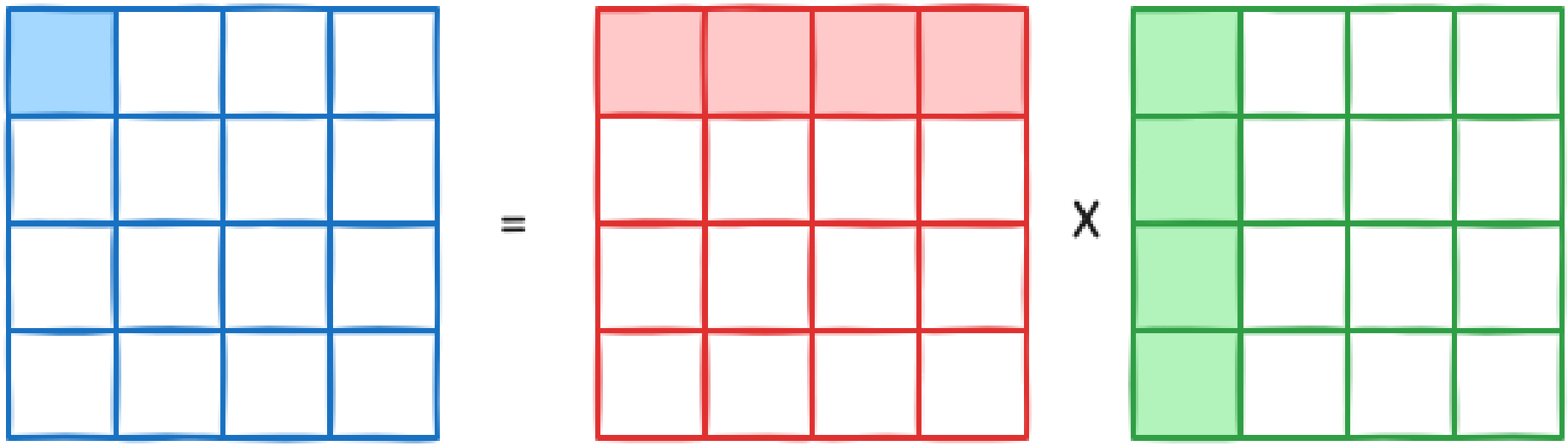
Graphics rendering is difficult because of the **volume** of similar computations, not because any one computation is especially complex.

Matrix Multiplication is Parallel



Each element of the output is independent. The results of C_{11} does not effect the results of C_{12}, or any other element.

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Key Idea

If matrix multiplication is a **parallel** task, and GPU's were designed for the **parallel** task of rendering graphics, can we repurpose GPU's to do matrix multiplication?

Takeaway

- CPUs can't efficiently perform matrix multiplication at scale.
- Graphics rendering is a parallel task, and so is matrix multiplication. Thus, GPUs can be good at matrix multiplication.
- GEMM is a generalization of of matrix multiplication, and is the operation we will be discussing.

Whats next

I've suggested that GPUs *can* do matrix multiplication, but how do we actually get there?